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Ch12 ACID (Introduction)

↳ *will cover in detail in Phase 22 (transactions) where we will dive deep into ACID, concurrency, isolation levels, MVCC, locks, deadlocks*

EG

Account	Balance
Arpit	250000
Aadya	125000

now let Arpit give his everything to Aadya
then
(s1) deduct 250k from Arpit
(s2) append 250k to Aadya

but if transaction fails after (s1) , database becomes inconsistent as money disappears
ACID prevents situation like this

ACID is a set of 4 properties that guarantees reliable, correct and safe execution of database transactions even in presence of failure or concurrent access

A → Atomicity **C → Consistency** **I → Isolation** **D → Durability**

Transaction

→ a sequence of one or more database operation treated as a single Indivisible operation

Atomicity

→ guarantees that either every operation in a transaction completes successfully or none of them do . no partial completion

Consistency

→ guarantees every committed transaction moves database from one valid state to another valid state while preserving all constraints and business rules

> Eg balance can't be -ve

Isolation

→ guarantees that concurrent transaction execute as if they were running one after other preventing unwanted performance interference

> Eg 2 people can't buy last product

Durability

→ guarantees that once transaction is committed , its changes will survive power failure , crashes or system restart.